

IT3051 – FUNDAMENTALS OF DATA MINING

MINI PROJECT ASSIGNMENT 2026



Dataset Validation Proposal

Group Number: 05 (Cognita)

Batch: Kandy Uni

Registration Number	Student Name
IT23547988	U.O.F DE RUN
IT23563032	THEAVER S.T.T
IT23721500	DILUKSHINI S
IT23620766	J.A.I.M.JAYAWARDHANA

Table of Contents

1. Introduction

2. Problem Scenario and Requirements

3. Dataset Identification

4. Dataset Suitability and Considerations

5. Proposed Project Direction

6. Conclusion

1. Introduction

1.1 Background

In manufacturing settings, failures of industrial equipment are a principal cause of unexpected downtime and higher operating costs. With conventional maintenance methods either a reactive approach (carrying out repairs after a failure has occurred) or a scheduled one (involving servicing at fixed intervals) there is either the possibility of sudden breakdowns or the need to carry out maintenance on equipment that hasn't yet failed. Predictive maintenance overcomes this problem by utilizing data from the operation of the machinery and from sensors to foresee failures before they take place, enabling maintenance to be planned according to the real condition of the equipment rather than on the basis of fixed timetables or repairs made after the fact.

1.2 Problem Statement

Industrial machines are liable to fail suddenly, causing expensive downtime, disrupting production schedules, and forcing reactive maintenance choices. At present, most monitoring methods do not provide an early, data driven way of working out which machines are in danger and what kind of failure might be developing. This project deals with the issue of using data from machine operation and from sensors to predict the diagnostic condition of industrial equipment, with the aim of detecting possible machine failures at an early stage and thus reducing both unexpected downtime and maintenance costs.

1.3 Project Objective

This project is intended to create a data mining and machine learning solution which predicts the diagnostic condition of industrial machinery covering normal operation as well as five specific types of failure by using the machine's operating and sensor measurements. The project seeks to show the entire data mining process, covering all the stages from problem understanding and dataset validation via exploratory data analysis, preprocessing, model development and optimization, to the production of a working predictive system that can be used by maintenance engineers, machine operators, and production managers.

2. Problem Scenario and Requirements

2.1 Prediction Objective

The focus of this project is predicting the diagnostic condition of industrial machinery using recorded operating and sensor data. Instead of only showing or flagging a generic failure, the system should be able to distinguish a normal operation from specific failure conditions. As the basis for this prediction, sensor readings such as air temperature, process temperature, rotational speed, torque and tool wear are being used. The aim is to detect failures before the actual machine breakdowns. Thus, the maintenance teams could move from reactive measures that are happening after the breakdown, to proactive measures that happen before the breakdown. By this they have the mitigate the risk of having to address the problem of unplanned downtime and high cost associated with the unexpected equipment failure.

2.2 Target Variable

The target variable selected for this project is Diagnostic, obtained from the AI4I-PMDI dataset. At the time of the observation of each it is a categorical variable that records the diagnostic condition of the machine. The variable contains six possible classes,

- No failure
- Heat Dissipation Failure
- Overstrain Failure
- Power Failure
- Tool Wear Failure
- Random Failures.

Unlike a simple binary flag that identifies only failure or no failure, this variable identifies the exact type of failure occurred if there is a failure. This helps maintenance engineers a clear idea instead of a simple yes or a no. When you refer the section 3.7, you will be able to see that the classes are heavily imbalanced, with 96.52% of records being labeled as 'No failure' and only 3.48% is spreading across the mentioned failure types. This imbalance is important as it has a direct implication for how the model should be trained and evaluated as discussed further in section 2.7.

2.3 Users and Stakeholders

The proposed predictive-maintenance system is intended to support personnel involved in the operation, monitoring, and maintenance of industrial equipment. The main users and stakeholders are maintenance engineers, factory or production managers, and machine operators.

- **Maintenance engineers** are the primary users of the system. They are responsible for investigating abnormal machine conditions, diagnosing possible failures, planning maintenance activities, and carrying out repairs. The system will provide them with the predicted diagnostic class and the operating or sensor values associated with the prediction. This information can help them prioritize inspections and determine whether further technical investigation is required.
- **Factory or production managers** are responsible for maintaining production continuity, managing downtime, and coordinating production and maintenance schedules. They can use summarized prediction results to identify machines that may require attention and to plan maintenance activities during suitable production intervals.
- **Machine operators** are responsible for operating industrial equipment and observing its day-to-day behavior. They can use the system to enter or review current machine measurements and identify whether the machine is predicted to be operating normally or associated with a possible failure condition. The output should be presented in a simple and understandable manner so that operators know when to inform maintenance personnel.
- Other stakeholders may include reliability engineers, supervisors, and information-technology personnel. Reliability engineers may use prediction results to investigate recurring failure patterns, while IT personnel may be responsible for maintaining the application, data storage, and model deployment.

The system is intended to support human decision-making rather than replace qualified maintenance personnel. Final maintenance and shutdown decisions should be confirmed by authorized technical staff.

2.4 Decisions Supported by the System

The system is designed to support maintenance and operational decisions based on the predicted diagnostic condition of industrial equipment.

The system can support the following decisions:

- Whether a machine should continue under normal monitoring or be referred for further inspection.
- Which machines should receive maintenance attention first.
- Whether a possible failure condition requires routine investigation or urgent technical attention.
- What type of preliminary investigation may be appropriate based on the predicted class, such as checking temperature, load, power-related conditions, or tool wear.
- Whether maintenance should be scheduled during a planned production break.
- Whether production managers should consider adjusting the production schedule to accommodate an inspection or maintenance activity.
- Whether additional sensor measurements should be collected before making a final maintenance decision.
- Whether the prediction should be escalated to a maintenance engineer for confirmation.

For example, if the system predicts **Tool Wear Failure**, the maintenance engineer may inspect the tool condition and consider tool replacement. If the system predicts **Heat Dissipation Failure**, the engineer may investigate the temperature measurements, cooling system, and ventilation conditions. If the system predicts **No failure**, the machine may continue under normal monitoring, subject to the organization's normal safety procedures.

The system will not automatically shut down machines, authorize repairs, or make final maintenance decisions. Its predictions will be used as decision-support information and must be verified by responsible personnel.

2.5 User Requirements

The proposed system should satisfy the following functional, usability, and data-processing requirements.

Functional Requirements

- The system should accept relevant machine operating and sensor measurements as input.
- The system should validate the entered values and identify missing, invalid, or out-of-range inputs.
- The system should preprocess input data using the same preprocessing pipeline used during model training.
- The system should predict one of the six diagnostic classes:
 - No failure.
 - Heat Dissipation Failure.
 - Overstrain Failure.
 - Power Failure.
 - Tool Wear Failure.
 - Random Failures.
- The system should display the predicted diagnostic class clearly.
- The system should provide a prediction confidence or probability where supported by the selected model.
- The system should provide a warning or recommendation to consult maintenance personnel when a failure class is predicted.
- The system should allow users to make a new prediction without restarting the application.

Data and Modelling Requirements

- The system should handle missing sensor values using an appropriate preprocessing method.
- The system should support categorical variables through suitable encoding.
- The system should handle numerical variables using appropriate scaling where required by the selected algorithm.
- The system should use the final trained model together with the correct preprocessing pipeline.
- The system should consider the strong class imbalance in the target variable.
- The model should be evaluated using suitable metrics such as precision, recall, F1-score, and a confusion matrix rather than accuracy alone.
- The system should avoid using identifiers as predictive features unless their use is justified.
- The system should provide reliable failure detection while recognizing that predictions may contain errors.

Usability Requirements

- The interface should be simple enough for machine operators and non-technical users.
- Input fields should include clear labels and measurement units.
- The system should display validation messages when users enter invalid values.
- The prediction result should be shown using understandable language.
- Failure predictions should be visually distinguishable from normal-operation predictions.
- The system should explain that a prediction is a decision-support indication and not a guaranteed diagnosis.
- The system should provide sufficient information for maintenance engineers to interpret the result.
- Role-specific dashboards and integration with industrial maintenance software may be considered as future improvements if they are outside the scope of the mini project.

2.6 Expected System Inputs and Outputs

Expected System Inputs

The system will receive the machine and sensor information required by the final selected model. The proposed input variables are shown below.

Input variable	Type	Description	Proposed use
Type	Categorical	Product or machine type	Model input after encoding
Air temperature (K)	Numerical	Air temperature measurement	Model input after missing-value treatment
Process temperature (K)	Numerical	Process temperature measurement	Model input after missing-value treatment
Rotational speed (rpm)	Numerical	Rotational speed of the machine	Model input after missing-value treatment
Torque (Nm)	Numerical	Torque applied during operation	Model input after missing-value treatment
Tool wear (min)	Numerical	Tool-wear measurement	Model input after missing-value treatment
Control	Categorical	Control-related information	Included only if data analysis confirms usefulness
System	Categorical or numerical	System-related information	Included only if justified after analysis

Date	Date/time	Date and time of the observation	May be transformed into time features if justified
------	-----------	----------------------------------	--

UDI and Product ID are identification fields. They should not normally be used directly as predictive inputs because they may not represent machine behavior. Their suitability will be examined during EDA and feature selection.

The final input variables may change after data-quality analysis, feature selection, and model experimentation. Any changes will be documented and justified in the later sections of the project.

Expected System Outputs

The system is expected to produce the following outputs:

Output	Type	Description
Predicted diagnostic class	Categorical	The predicted machine condition from the six target classes
Prediction confidence	Numerical	The model's confidence or probability for the predicted class, if available
Failure status	Categorical	Normal operation or possible failure condition
Input validation message	Text	Notification if an input is missing, invalid, or outside an acceptable range
Basic maintenance guidance	Text	A general recommendation to continue monitoring or refer the machine to maintenance personnel

The primary output will be the predicted diagnostic class. For example, the system may display:

Predicted condition: Heat Dissipation Failure

Confidence: 0.81

Suggested action: Refer the machine to a maintenance engineer for inspection of temperature and cooling-related conditions.

The system will not claim that a machine will definitely fail, and it will not automatically authorize a shutdown or repair. The output will be presented as a prediction to support further investigation and maintenance planning.

2.7 Data Mining Task

This project is framed as a **supervised classification task**. Since the target variable consists of six discrete categories than just continuous value range, the appropriate task would be multiclass classification rather than regression. The input features include air temperature, process temperature, rotational speed, torque, tool wear, and machine type. The goal of the model is to be

able to predict from the six diagnostic classes which one best matches the given observation. Because of the class, imbalance accuracy alone is not sufficient to measure performance. A model can be highly accurate by simply predicting 'No failure' for almost every case while being unable to detect real faults completely. For this reason, precision, recall, F1 score and a per class confusion matrix will be used to properly evaluate performance. Techniques such as stratified train/test splitting, class-weighted models, or resampling methods like SMOTE may be applied to address this imbalance. These steps will also be included alongside the missing values handling strategies described in the upcoming sections.

3. Dataset Information

3.1 Dataset Name

The dataset selected for this project is the AI4I-PMDI: Predictive Maintenance Dataset with Irregularities. AI4I-PMDI is a predictive maintenance dataset designed to represent more realistic industrial data conditions by incorporating irregularities such as missing data and irregular time-series observations. The dataset is based on the AI4I 2020 Predictive Maintenance Dataset.

3.2 Original Source and Dataset URL

The dataset was obtained from Figshare, a publicly accessible research data repository. It was authored by Jean-Victor Autran and published on 13 June 2024. The dataset is available through the following source:

AI4I-PMDI Dataset – Figshare

The dataset has the persistent identifier DOI: 10.6084/m9.figshare.24718428.

The dataset is associated with the research work “AI4I-PMDI: Predictive maintenance datasets with complex industrial settings' irregularities”, published in *Procedia Computer Science*, Volume 246, 2024, pages 1201–1209.

3.3 Dataset Citation

The dataset can be cited as:

Autran, J.-V. (2024). AI4I-PMDI: Predictive Maintenance Dataset with Irregularities. Figshare. <https://doi.org/10.6084/m9.figshare.24718428>

The research publication associated with the dataset can additionally be cited as:

Autran, J.-V., Kuhn, V., Diguet, J.-P., Dubois, M., & Buche, C. (2024). AI4I-PMDI: Predictive maintenance datasets with complex industrial settings' irregularities. *Procedia Computer Science*, 246, 1201–1209. <https://doi.org/10.1016/j.procs.2024.09.546>

3.4 Dataset Description

AI4I-PMDI is a predictive maintenance dataset containing information related to the operating conditions and diagnostic state of industrial equipment. It is derived from the AI4I 2020 predictive maintenance dataset and introduces data irregularities intended to better represent challenges encountered in industrial environments, particularly missing data and irregular time-series data.

The dataset contains machine identification information, product-related information, operating conditions, and sensor measurements. The relevant variables identified for this project include:

Feature	Description / Role
UDI	Unique identification value for an observation
Date	Date/time information associated with the observation
System	System-related identifier/information
Control	Control-related information
Product ID	Product identification information
Type	Product/machine type information
Air temperature (K)	Air temperature measurement in Kelvin
Process temperature (K)	Process temperature measurement in Kelvin
Rotational speed (rpm)	Rotational speed of the equipment
Torque (Nm)	Torque measurement of the equipment
Tool wear (min)	Tool wear measurement in minutes
Diagnostic	Diagnostic condition of the equipment and the project's target variable

These variables provide information about machine operating conditions that can be investigated for their relationship with equipment diagnostic conditions.

3.5 Real-World Context

The dataset is designed for the predictive maintenance (PdM) domain. Predictive maintenance uses equipment and sensor data to identify equipment conditions and potential failures before they result in unexpected downtime.

The original AI4I-PMDI research highlights that publicly available predictive-maintenance datasets can lack realistic industrial complexities. AI4I-PMDI was introduced to incorporate irregularities such as missing measurements and irregular data acquisition, making the dataset more representative of challenges that can occur in complex industrial settings.

This real-world context directly supports the proposed project because industrial organizations can use operating and sensor measurements to monitor equipment conditions and support maintenance-related decision-making.

3.6 Dataset Size and Relevant Features

The dataset contains 10,000 records. The dataset used in this project contains the following 12 columns:

1. UDI
2. Date
3. System
4. Control
5. Product ID
6. Type
7. Air temperature (K)
8. Process temperature (K)
9. Rotational speed (rpm)
10. Torque (Nm)
11. Tool wear (min)
12. Diagnostic

The features can be broadly categorized as:

- **Identification/information features:** UDI, Date, Product ID, System
- **Categorical/operational features:** Control, Type
- **Numerical operating/sensor features:** Air temperature, Process temperature, Rotational speed, Torque, Tool wear
- **Target variable:** Diagnostic

The dataset's inclusion of missing and irregular observations is particularly relevant to this project because it allows the group to investigate realistic data-quality challenges during the later EDA and preprocessing stages.

3.7 Target Variable and Its Meaning

The target variable in the selected dataset is Diagnostic. It represents the diagnostic condition of the industrial machine for each recorded observation. Unlike a simple binary machine-failure indicator, the selected AI4I-PMDI dataset provides a diagnostic category indicating whether the machine is operating normally or which type of failure has been identified.

The dataset contains six diagnostic classes:

Diagnostic Class	Meaning
No failure	The machine is operating without a recorded failure condition.
Heat Dissipation Failure	The machine has experienced a failure associated with insufficient heat dissipation under the conditions represented in the dataset.
Overstrain Failure	The machine has experienced a failure associated with excessive mechanical load/strain.
Power Failure	The machine has experienced a failure related to the power conditions of the process.
Tool Wear Failure	The machine has experienced a failure associated with excessive tool wear.
Random Failures	The machine has experienced a randomly occurring failure that is not directly determined by the main operating parameters.

The original AI4I dataset describes these failure mechanisms as tool wear failure (TWF), heat dissipation failure (HDF), power failure (PWF), overstrain failure (OSF), and random failure (RNF). The original dataset also explains that a machine-failure label can result when at least one of these failure conditions occurs.

For this project, the Diagnostic variable can therefore be treated as a multiclass classification target. The machine operating conditions and sensor measurements can be used as input features, while the model attempts to predict the diagnostic condition.

This target directly supports the project's objective of identifying potential machine failures early. Instead of only predicting whether a machine will fail, the system can potentially provide information about what type of diagnostic condition is associated with the observation.

Distribution of the target variable

An inspection of the provided CSV shows the following distribution:

Diagnostic	Number of Records	Percentage
No failure	9,652	96.52%
Heat Dissipation Failure	106	1.06%
Overstrain Failure	98	0.98%
Power Failure	83	0.83%
Tool Wear Failure	42	0.42%
Random Failures	19	0.19%
Total	10,000	100%

The distribution shows a strong class imbalance, because the majority of observations represent normal machine operation while the different failure categories represent relatively small portions of the dataset. This is an important consideration for the later machine-learning stage. A model could obtain high overall accuracy by mainly predicting the majority of class while performing poorly on the less frequent failure classes.

Therefore, evaluation should not rely only on accuracy. Metrics such as precision, recall, F1-score and a confusion matrix should also be considered, particularly for the individual failure classes.

4. Dataset Suitability and Considerations

4.1 Suitability for the Proposed Problem

The AI4I-PMDI dataset is suitable for the proposed predictive-maintenance problem because it contains machine operating measurements together with diagnostic information that can be used to develop a machine-learning model for failure detection and diagnosis.

The original AI4I dataset was specifically created to represent predictive-maintenance data encountered in industrial environments. It contains machine/process measurements such as air temperature, process temperature, rotational speed, torque, and tool wear, together with machine-failure information.

The selected AI4I-PMDI version extends to the original AI4I dataset by introducing missing data and irregular time-series characteristics. The dataset authors specifically describe it as being based on AI4I 2020 but modified to contain missing data and irregular time-series data.

This makes the dataset particularly relevant to the proposed project because real industrial monitoring systems may not always produce perfectly complete or regularly sampled sensor observations.

Reasons the dataset is suitable

1. Relevant machine-operating features

The dataset contains several variables representing machine and process conditions, including:

- Air temperature
- Process temperature
- Rotational speed
- Torque
- Tool wear
- Product type
- System/control information

These variables provide information that can potentially be used to distinguish normal operation from different machine-failure conditions.

2. Appropriate target variable

The Diagnostic variable provides six diagnostic categories. This makes the dataset suitable for a multiclass classification problem, where the objective is to predict the diagnostic condition from available machine measurements.

3. Contains failure examples

The dataset does not contain only normal machine-operation records. It contains examples of several failure types, allowing a model to learn about relationships between operating conditions and diagnostic outcomes.

However, the number of observations for some failure classes is relatively small. This needs to be considered during model training and evaluation.

4. Realistic data-quality challenges

The AI4I-PMDDI dataset deliberately includes missing and irregular data.

This is valuable for a data-mining project because the project can demonstrate data-preprocessing techniques such as:

- Missing-value analysis
- Missing-value imputation
- Feature preprocessing
- Categorical encoding
- Handling irregular observations
- Class-imbalance handling
- Model evaluation under imperfect data conditions

5. Appropriate dataset size

The uploaded dataset contains **10,000 observations**, which is large enough for a university-level machine-learning project involving exploratory analysis, preprocessing, model training, and evaluation.

6. Direct connection to the proposed users

The dataset supports a system that could provide diagnostic information to maintenance engineers, production managers, and machine operators. For example, a predicted failure condition could be used as an input to a maintenance investigation or operational decision.

Overall suitability

The dataset provides an appropriate foundation for the proposed project, but it should not be interpreted as a complete representation of every real industrial environment. It is best viewed as a research/educational dataset for demonstrating predictive-maintenance techniques, rather than sufficient evidence for deploying a safety-critical industrial maintenance system.

4.2 Expected Machine Learning Task

The multiclass classification task described in Section 2.7 is directly supported by the A14I-PMDI. The six category diagnostic labels and the relevant sensor and operating features provided in the dataset form a suitable basis for this prediction task. When we consider the presence of several distinct failure categories, rather than a single failure flag, it shows that a multiclass approach is more appropriate than a simple binary classification. However, achieving reliable performance will require addressing the class imbalance (Section 3.7) and substantial missing values (Section 4.3) identified during initial data inspection. Overall, the task is achievable using this dataset, provided appropriate preprocessing is carried out beforehand.

4.3 Initial Data-Quality Observations

An initial inspection was performed on the uploaded CSV file.

The dataset contains:

- **10,000 rows**
- **12 columns**
- No missing values in the identifier and categorical variables
- Substantial missing values in several sensor variables
- Six diagnostic classes
- No duplicate complete rows detected
- No duplicate UDI values detected

4.3.1 Dataset structure

The uploaded dataset contains the following variables:

Variable	Type	Initial observation
----------	------	---------------------

UDI	Integer	Unique identifier
Date	Date/time	Observation timestamp
System	Integer	System-related identifier/category
Control	Categorical	Three categories
Product ID	Categorical	Unique product identifier
Type	Categorical	Product type
Air temperature (K)	Numerical	Sensor measurement
Process temperature (K)	Numerical	Sensor measurement
Rotational speed (rpm)	Numerical	Machine operating measurement
Torque (Nm)	Numerical	Machine operating measurement
Tool wear (min)	Numerical	Tool-wear measurement
Diagnostic	Categorical target	Machine diagnostic condition

4.3.2 Missing values

The most significant data-quality issue is the presence of missing sensor measurements.

The following missing-value rates were observed in the uploaded file:

Feature	Missing Values	Percentage
Air temperature	6,563	65.63%
Process temperature	6,563	65.63%
Rotational speed	3,321	33.21%
Torque	3,437	34.37%
Tool wear	6,679	66.79%
UDI	0	0%
Date	0	0%
System	0	0%
Control	0	0%
Product ID	0	0%
Type	0	0%
Diagnostic	0	0%

The missingness is therefore substantial rather than negligible. **Air temperature, process temperature, and tool wear have missing values in more than 60% of the records.**

This is an important issue because these variables are potentially useful for predictors of machine conditions. Removing every row containing a missing sensor value could result in a substantial loss of training data.

Consequently, the preprocessing stage should investigate appropriate missing-value strategies rather than simply deleting all incomplete observations.

4.3.3 Class imbalance

Another major data-quality consideration is the imbalance in the target variable.

There are **9,652 normal observations compared with only 348 failure observations** across all failure types.

The smallest class, Random Failures, contains only **19 observations**, while Tool Wear Failure contains 42 observations.

This imbalance can affect model training because a model may learn to favour the majority of class. Therefore, the project should consider techniques such as:

- Stratified train/test splitting
- Class-weighted models
- Appropriate resampling techniques
- Precision, recall and F1-score
- Confusion matrix analysis
- Per-class performance analysis

4.3.4 Duplicate records

The initial inspection found:

- **0 complete duplicate rows**
- **0 duplicate UDI values**
- **0 duplicate Product ID values**

Therefore, there is no immediate indication that duplicated observations need to be removed.

However, identifiers such as UDI and Product ID should be examined carefully before model training because identifiers generally should not be used directly as predictive features unless there is a justified reason.

4.3.5 Numerical feature ranges

The sensor measurements also have different units and scales. For example:

- Temperature is measured in Kelvin.
- Rotational speed is measured in rpm.
- Torque is measured in Nm.
- Tool wear is measured in minutes.

Depending on the selected machine-learning algorithms, feature scaling or normalization may therefore be required.

4.3.6 Data-quality preprocessing requirements

Based on the initial inspection, the following preprocessing activities are recommended:

1. Check the structure and data types.
2. Investigate missing-value patterns.
3. Determine an appropriate missing value treatment.
4. Examine numerical-feature distributions and potential outliers.
5. Encode categorical variables where required.
6. Evaluate whether identifier fields should be excluded.
7. Examine the class imbalance.
8. Use stratified sampling when creating training and testing sets.
9. Apply feature scaling where required by the selected algorithm.
10. Check for potential data leakage before model training.

4.4 Known Limitations

Although the dataset is suitable for the proposed academic project, several limitations should be acknowledged.

4.4.1 Synthetic nature of the underlying dataset

The original AI4I 2020 dataset is a synthetic dataset designed to reflect predictive-maintenance data encountered in industry, rather than a collection of directly observed production-machine data.

Therefore, patterns learned from this dataset may not necessarily represent all real industrial machines, operating environments, or failure mechanisms.

The resulting model should consequently be considered a demonstration or research model rather than a production-ready maintenance system.

4.4.2 Severe class imbalance

Failure conditions are much less frequent than normal operations.

For example, the uploaded dataset contains:

- 9,652 normal observations
- 106 heat-dissipation failures
- 98 overstrain failures
- 83 power failures
- 42 tool-wear failures
- 19 random failures

The limited number of examples for some failure types may make it difficult for a model to learn reliable patterns for those classes.

4.4.3 Large amount of missing sensor data

The selected AI4I-PMDI version contains missing information.

While this provides a useful preprocessing challenge, it also reduces the amount of complete sensor information available for prediction.

For example, approximately 66.79% of Tool Wear measurements are missing in the provided CSV.

4.4.4 Limited number of features

The dataset contains a relatively small number of machine measurements compared with the range of sensors that may be available in a real industrial predictive-maintenance system.

Real industrial systems may include:

- Vibration sensors
- Acoustic sensors
- Pressure sensors
- Current/voltage measurements
- Lubrication measurements
- Motor temperature
- Historical maintenance records
- Environmental conditions

These are not all represented in the selected dataset.

4.4.5 Limited failure representation

The dataset contains only a limited set of predefined failure categories. Real machines can experience many other types of faults, and multiple failure mechanisms can potentially occur simultaneously.

The original AI4I documentation also notes that the machine-failure label does not make the underlying failure mechanism fully transparent when multiple failure conditions may be involved.

4.4.6 Generalization limitation

A model trained on this dataset may perform well on similar data but may not necessarily generalize to:

- A different machine manufacturer
- Different factory conditions
- Different sensor configurations
- Different operating ranges
- Different maintenance procedures

Therefore, external validation using real industrial data would be necessary before practical deployment.

4.4.7 Prediction does not replace maintenance expertise

A machine-learning prediction should be treated as a decision-support signal rather than an automatic replacement for qualified maintenance procedures.

For an actual industrial deployment, predictions would need to be validated against machine specifications, maintenance history, and expert knowledge.

4.5 Ethical and Privacy Considerations

The dataset does not appear to contain direct personal information such as names, addresses, telephone numbers, or medical information. Its variables primarily describe machine operation, sensor measurements, product information, and diagnostic conditions.

Therefore, personal privacy risk is relatively low compared with datasets containing human subjects.

However, several ethical considerations remain relevant.

4.5.1 Responsible use of predictions

A predictive-maintenance system can influence decisions such as:

- When to inspect a machine
- Whether to schedule maintenance
- Whether to temporarily stop equipment
- Which machine requires attention first

Incorrect predictions could potentially result in unnecessary maintenance or, more seriously, failure to identify a genuine machine problem.

Therefore, predictions should be presented as decision-support information, particularly during the early stages of system deployment.

4.5.2 False negatives

A false negative occurs when the system predicts normal operation when a machine has a failure of condition.

In predictive maintenance, this can be important because failure to identify a problem may contribute to unexpected downtime or equipment damage.

Therefore, the project should evaluate recall for individual failure classes rather than relying only on overall accuracy.

4.5.3 False positives

A false positive occurs when the system predicts a failure when the machine is actually operating normally.

Although this may be less serious than some false-negative situations, excessive false positives could result in:

- Unnecessary inspections
- Unnecessary maintenance
- Production interruptions
- Increased operational costs

The final system should therefore consider both types of errors.

4.5.4 Transparency

For maintenance-related decision support, users should ideally understand why a prediction was produced.

Where possible, explainability techniques could be considered to identify which features contributed to a prediction. This can help maintenance engineers interpret the output rather than treating the machine-learning model as a completely unexplained decisionmaker.

4.5.5 Dataset privacy and organizational information

Although the provided dataset itself does not contain obvious personal information, real industrial predictive-maintenance datasets may contain commercially sensitive information.

For example, real datasets could reveal:

- Production volumes
- Equipment performance
- Failure rates
- Maintenance schedules
- Factory operating conditions
- Proprietary process information

If the proposed system is later extended to real company data, access control, anonymization and appropriate data-governance procedures would be required.

4.6 Licensing and Accessibility

The selected dataset, AI4I-PMDDI: Predictive Maintenance Dataset with Irregularities, is publicly available through Figshare.

The dataset is authored by Jean-Victor Autran and was published on Figshare in 2024. The dataset page identifies the dataset as AI4I-PMDDI, based on the AI4I 2020 dataset and modified to include missing data and irregular time-series data.

The dataset is distributed under the Creative Commons Attribution 4.0 International (CC BY 4.0) license.

The CC BY 4.0 license generally permits users to share and adapt the material, provided that appropriate attribution is given.

For the project report, the dataset should therefore be properly acknowledged and cited. The original AI4I 2020 dataset should also be acknowledged because AI4I-PMDI is based on it. The original AI4I 2020 dataset is available through the UCI Machine Learning Repository and is also licensed under CC BY 4.0.

Dataset source: [AI4I-PMDI — Figshare](#)

Underlying AI4I 2020 dataset: UCI Machine Learning Repository — AI4I 2020 Predictive Maintenance Dataset

Suggested attribution

For the report, the dataset can be acknowledged approximately as:

Autran, J.-V. (2024). *AI4I-PMDI: Predictive Maintenance Dataset with Irregularities*. Figshare. Licensed under CC BY 4.0. The dataset is based on the AI4I 2020 Predictive Maintenance Dataset by S. Matzka.

The original AI4I paper is S. Matzka, “**Explainable Artificial Intelligence for Predictive Maintenance Applications,**” **2020 Third International Conference on Artificial Intelligence for Industries (AI4I)**, pp. 69–74. This publication is identified by both the AI4I-PMDI page and the UCI dataset documentation.

5. Proposed Project Direction

5.1 Proposed Data Mining Approach

The proposed project will follow a complete data-mining workflow consisting of data understanding, preprocessing, exploratory data analysis, feature engineering, model development, evaluation, and deployment as a user-facing predictive system. The objective is not simply to apply machine-learning algorithms, but to investigate how data-quality issues, class imbalance, feature characteristics, and different modelling techniques affect the prediction of industrial machine diagnostic conditions.

The proposed workflow is shown below:

Dataset → Data Preprocessing → Exploratory Data Analysis → Feature Engineering → Train/Test Split → Model Training → Model Evaluation → Model Selection → Predictive System

Step 1: Data preprocessing

The first stage will involve validating the dataset structure and preparing the data for modelling. The following activities are planned:

- Checking data types and converting the Date field into an appropriate date/time format.
- Identifying missing values and examining their distribution across features.
- Investigating whether missing values occur randomly or show particular patterns.
- Applying suitable imputation techniques to numerical sensor variables where appropriate.
- Evaluating whether records with excessive missing information should be retained, transformed, or removed based on evidence from the analysis.
- Encoding categorical variables such as Type and other selected categorical features.
- Examining UDI and Product ID as identifiers and excluding them from modelling if they do not provide meaningful predictive information.
- Investigating System and Control to determine whether they contain useful information for prediction.
- Checking for duplicate records and inconsistent values.
- Detecting potential outliers in numerical sensor measurements.
- Checking for data leakage before model training.

Because the selected AI4I-PMDI dataset deliberately introduces missing and irregular observations, missing-data treatment will be an important part of the project rather than simply removing incomplete records. The dataset authors describe AI4I-PMDI as a modified version of AI4I 2020 containing missing data and irregular time-series data.

Step 2: Exploratory Data Analysis

Exploratory Data Analysis (EDA) will be performed to understand relationships between machine operating conditions and diagnostic classes. The analysis will include:

- Frequency distributions of the six diagnostic classes.
- Histograms and boxplots for numerical variables.
- Analysis of missing-value patterns.
- Correlation analysis between numerical features.
- Comparison of sensor measurements across diagnostic classes.
- Analysis of temperature differences between process and air temperature.
- Investigation of relationships between rotational speed, torque, and machine condition.
- Analysis of tool wear in relation to failure conditions.
- Examination of categorical variables and their relationship with the target.

- Identification of possible outliers and unusual operating conditions.

The results of EDA will be used to justify feature-selection and feature-engineering decisions rather than selecting features arbitrarily.

Step 3: Feature engineering

Where supported by the data and domain knowledge, additional features may be derived from the existing variables. Potential engineered features include:

- **Temperature difference:** Process Temperature – Air Temperature.
- **Mechanical power:** Torque $\times$ rotational speed converted into an appropriate power-related measurement.
- **Load-related indicators:** combinations of torque and rotational speed.
- **Tool-load interaction:** Tool Wear $\times$ Torque, where appropriate.
- **Date/time-derived features:** such as day, hour, or time-related characteristics if the timestamp provides useful predictive information.

Feature engineering will only be applied where the derived variable has a logical interpretation and improves or contributes meaningfully to the prediction task. Any feature that creates target leakage or uses information that would not be available at prediction time will be excluded.

Step 4: Handling class imbalance

The target distribution contains a very large majority class and several rare failure classes. Therefore, the project will compare appropriate imbalance-handling strategies rather than relying on accuracy alone.

The following approaches may be investigated:

- Stratified train/test splitting.
- Class-weighted machine-learning algorithms.
- Random oversampling where appropriate.
- SMOTE or another suitable oversampling technique where the class sizes permit its use.
- Comparison of models with and without imbalance handling.

Any resampling technique will be applied only to the training data and, where cross-validation is used, within the training folds. This is necessary to avoid information leakage from synthetic or duplicated samples into validation data.

The extremely small number of observations in some failure classes means that the project will explicitly examine per-class performance. Recent research using the AI4I dataset also demonstrates that rare failure classes can remain difficult to detect even when overall model performance appears strong.

Step 5: Model development

Several classification algorithms will be implemented and compared. The initial candidate models may include:

1. **Logistic Regression** – as an interpretable baseline classification model.
2. **Decision Tree** – to provide an interpretable rule-based model.
3. **Random Forest** – to model nonlinear relationships and interactions between features.
4. **Support Vector Machine (SVM)** – as an alternative classification approach.
5. **Gradient Boosting / XGBoost** – as a more advanced tree-based approach.

The final model will not be selected in advance. Instead, the candidate models will be compared using the same preprocessing and evaluation framework. This will allow the project to determine which approach provides an appropriate balance between predictive performance, minority-class detection, interpretability, and computational requirements.

Step 6: Model evaluation

Because the dataset is highly imbalanced, accuracy will not be used as the only evaluation measure.

The following metrics will be considered:

- **Precision**
- **Recall**
- **F1-score**
- **Macro-averaged F1-score**
- **Weighted F1-score**
- **Confusion matrix**
- **Per-class precision, recall, and F1-score**

Recall will receive particular attention for failure classes because failing to identify an actual failure condition can be more important than correctly identifying the already dominant normal class.

The confusion matrix will also be used to determine which failure types are frequently confused with one another. This will provide more useful information than a single overall accuracy value.

If probability outputs are available from the selected model, confidence or probability estimates may also be examined. These values will be presented as model estimates rather than guarantees that a failure will occur.

Step 7: Model selection and optimization

After the initial comparison, the most suitable candidate models will be further investigated through hyperparameter tuning. Techniques such as grid search or randomized search may be used together with stratified cross-validation.

The final model will be selected based on the complete evaluation results rather than accuracy alone. Particular attention will be given to minority-class recall and macro-F1 because the project is intended to identify different failure conditions rather than simply predict the majority class.

Step 8: Explainability

Where practical, model-explanation techniques may be incorporated into the final system. For example, feature-importance methods or SHAP-based explanations could be investigated to identify which operating measurements contributed most strongly to an individual prediction.

This is particularly useful in predictive maintenance because maintenance personnel should be able to interpret the prediction as supporting information rather than treating the model as an unexplained decision-maker.

Step 9: Data-mining system implementation

The final selected model and preprocessing pipeline will be integrated into a simple user-facing application.

A user will enter the available machine measurements, including relevant temperature, rotational speed, torque, tool-wear, and categorical information. The system will validate the inputs, apply the same preprocessing used during model training, and return the predicted diagnostic condition.

The proposed workflow is:

User Input → Input Validation → Preprocessing → Trained Classification Model → Diagnostic Prediction → Confidence/Probability → Maintenance Decision-Support Information

The system will not automatically shut down equipment or authorize maintenance. The prediction will be provided as decision-support information for qualified personnel.

5.2 Expected Outcome

The expected outcome of the project is a working machine-learning-based predictive-maintenance prototype capable of classifying an industrial machine observation into one of the six diagnostic categories:

- No Failure
- Heat Dissipation Failure
- Overstrain Failure
- Power Failure
- Tool Wear Failure
- Random Failures

The project is expected to produce the following outcomes:

1. A cleaned and analysed dataset

The project will produce a documented preprocessing pipeline that addresses the major data-quality issues identified during the initial analysis, particularly missing sensor measurements, categorical variables, potential outliers, and class imbalance.

2. Identification of useful predictive features

EDA and feature-selection analysis are expected to identify which machine operating variables provide useful information for distinguishing between normal operation and different failure conditions.

3. Comparison of machine-learning models

Multiple classification algorithms will be trained and evaluated using a consistent experimental procedure. Their performance will be compared using precision, recall, F1-score, macro-F1, and confusion matrices.

4. A selected predictive model

The project will identify a final model based on the experimental results. The selected model should provide useful performance across the minority failure classes rather than obtaining a high score mainly through prediction of the majority No Failure class.

5. A predictive-maintenance application

A user-friendly prototype will be developed so that a user can enter machine measurements and receive a predicted diagnostic condition.

6. Decision-support output

The system will provide an understandable result such as:

- **Predicted condition:** Tool Wear Failure
- **Prediction probability:** [model-generated value]
- **Status:** Possible failure condition
- **Suggested action:** Refer the machine to qualified maintenance personnel for further inspection.

The suggested action will remain general and will not replace professional maintenance procedures.

7. Demonstration of a complete data-mining solution

The final project will demonstrate the application of data mining to a real-world-oriented industrial problem by connecting data preprocessing, exploratory analysis, feature engineering, classification, evaluation, and deployment into one system.

The expected result is therefore not simply a trained machine-learning model. It is a complete decision-support prototype that demonstrates how machine and sensor data can be transformed into useful diagnostic information.

The results should nevertheless be interpreted within the limitations of the dataset. AI4I-PMDI is based on the synthetic AI4I 2020 dataset, and therefore successful performance on this dataset does not establish that the model would perform equally well on real production equipment.

5.3 Proposed System Overview

The proposed system will consist of five main stages: **Input, Validation and Preprocessing, Prediction, Interpretation, and Decision Support.**

5.3.1 System workflow

The proposed system will operate according to the following workflow:

Step 1 – User Input

The user enters the available machine information through the application interface. The inputs may include:

- Machine/Product Type
- Air Temperature

- Process Temperature
- Rotational Speed
- Torque
- Tool Wear
- Other selected operational variables supported by the final model

Step 2 – Input Validation

The system checks whether:

- Required fields have been provided.
- Numerical values are valid.
- Measurement units are correct.
- Values fall within acceptable ranges established during the data-analysis stage.
- Categorical values belong to the permitted categories.

Invalid inputs will generate an appropriate validation message instead of being passed directly to the model.

Step 3 – Preprocessing

The validated input will pass through the same preprocessing pipeline used during model development.

This may include:

- Missing-value treatment.
- Categorical encoding.
- Numerical feature scaling where required.
- Feature engineering.
- Feature selection.

Using the same preprocessing pipeline for training and prediction helps maintain consistency between the data used to develop the model and new user inputs.

Step 4 – Machine-Learning Prediction

The processed input will be passed to the final trained classification model.

The model will produce one of the six diagnostic classes:

No Failure | Heat Dissipation Failure | Overstrain Failure | Power Failure | Tool Wear Failure | Random Failures

Where supported by the selected model, the system will also display a probability or confidence estimate.

Step 5 – Result Interpretation

The prediction will be converted into an understandable system output.

For example:

Input:

- Air Temperature = 302 K
- Process Temperature = 311 K
- Rotational Speed = 1350 rpm
- Torque = 55 Nm
- Tool Wear = 210 min
- Type = M

Output:

Predicted condition: Heat Dissipation Failure

Status: Possible failure condition

Model probability: [calculated probability]

Suggested action: Refer the equipment to maintenance personnel for further inspection of temperature and cooling-related conditions.

The values above are an example of the system format and are not intended to represent an actual prediction before the final model is trained.

Step 6 – Decision Support

The system will provide general decision-support guidance based on the predicted class. For example:

Predicted condition	Example decision-support information
No Failure	Continue normal monitoring according to existing procedures.
Heat Dissipation Failure	Inspect temperature and cooling-related conditions.
Overstrain Failure	Investigate load, torque, and tool operating conditions.
Power Failure	Inspect power-related and operating conditions.
Tool Wear Failure	Inspect tool condition and consider maintenance according to technical procedures.
Random Failures	Escalate for further technical inspection because the failure is not directly explained by the main operating measurements.

These recommendations are intended only as general decision-support information. The final maintenance decision will remain with qualified personnel.

5.3.2 Proposed architecture

The proposed system can be represented as:

User Interface



Input Validation



Preprocessing Pipeline



Feature Engineering / Feature Selection



Trained Classification Model



Diagnostic Prediction



Confidence + Explanation



Maintenance Decision Support

The system will maintain a clear separation between the machine-learning prediction and the final maintenance decision. This is important because the dataset is synthetic and the model may produce incorrect predictions, particularly for the rare failure classes.

The final implementation may use a lightweight web-based interface connected to a Python machine-learning backend. The exact technologies will be selected during implementation according to the team's development requirements.

6. Conclusion

This project proposes a data-mining and machine-learning solution for the classification of industrial machine diagnostic conditions using the AI4I-PMDI predictive-maintenance dataset. The dataset is particularly suitable for the proposed academic project because it combines machine operating measurements with failure-related information and introduces data-quality challenges such as missing and irregular observations. The dataset is publicly available through Figshare under a CC BY 4.0 licence, while its underlying AI4I 2020 dataset is available through the UCI Machine Learning Repository.

The proposed problem is formulated as a supervised multiclass classification task. The system will attempt to distinguish between normal operation and five failure categories: Heat Dissipation Failure, Overstrain Failure, Power Failure, Tool Wear Failure, and Random Failures.

The initial dataset investigation has identified two major challenges: substantial missing sensor measurements and severe class imbalance. These challenges provide opportunities to demonstrate meaningful data-mining techniques rather than simply applying classification algorithms to a clean dataset. The project will therefore investigate missing-value treatment, categorical encoding, feature engineering, feature selection, class-imbalance handling, and leakage-free model evaluation.

Several machine-learning algorithms will be compared, including interpretable baseline models and tree-based ensemble methods. The final model will be selected based on a comprehensive evaluation using precision, recall, F1-score, macro-F1, and confusion-matrix analysis. Particular attention will be given to the minority failure classes because high overall accuracy could otherwise hide poor failure detection.

The expected final outcome is a working predictive-maintenance prototype through which users can enter machine operating information and obtain a predicted diagnostic condition together with supporting information. The system is intended to assist maintenance personnel rather than replace their technical judgment or automatically control industrial equipment.

Finally, the limitations of the dataset will be considered when interpreting the results. AI4I 2020 is explicitly described by UCI as a synthetic dataset designed to reflect predictive-maintenance data encountered in industry, while AI4I-PMDI modifies it by introducing missing and irregular observations. Therefore, the resulting model will be considered an academic decision-support prototype rather than a production-ready industrial safety system. Validation using real industrial data would be required before deployment in an actual manufacturing environment.